

## SAFETY DATA SHEET

Creation Date 06-Nov-2010

Revision Date 26-Mar-2024

Revision Number 3

### 1. Identification

**Product Name** Zirconium(IV) chloride

**Cat No. :** L14891

**CAS No** 10026-11-6  
**Synonyms** Zirconium tetrachloride

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
**Emergency Number US:**001-201-796-7100 / **Europe:** +32 14 57 52 99  
**CHEMTREC** Tel. No. **US:**001-800-424-9300 / **Europe:**001-703-527-3887

### 2. Hazard(s) identification

#### **Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                   |            |
|-----------------------------------|------------|
| Corrosive to metals               | Category 1 |
| Acute oral toxicity               | Category 4 |
| Skin Corrosion/Irritation         | Category 1 |
| Serious Eye Damage/Eye Irritation | Category 1 |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

May be corrosive to metals  
Harmful if swallowed  
Causes severe skin burns and eye damage

**Precautionary Statements****Prevention**

Keep only in original container

Wash face, hands and any exposed skin thoroughly after handling

Do not eat, drink or smoke when using this product

Wear protective gloves/protective clothing/eye protection/face protection

Wear respiratory protection

**Response**

Call a POISON CENTER or doctor/physician if you feel unwell

**Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

Immediately call a POISON CENTER or doctor/physician

**Skin**

Immediately call a POISON CENTER or doctor/physician

Wash contaminated clothing before reuse

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

Immediately call a POISON CENTER or doctor/physician

**Ingestion**

IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell

Rinse mouth

Do NOT induce vomiting

**Spills**

Absorb spillage to prevent material damage

**Storage**

Store locked up

Store in a well-ventilated place. Keep container tightly closed

Store in corrosive resistant polypropylene container with a resistant inliner

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

Reacts violently with water

Corrosive to the respiratory tract

### 3. Composition/Information on Ingredients

| Component               | CAS No     | Weight % |
|-------------------------|------------|----------|
| Zirconium tetrachloride | 10026-11-6 | >95      |

### 4. First-aid measures

**General Advice**

Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.

**Eye Contact**

Immediate medical attention is required. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical

|                                            |                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | attention is required.                                                                                                                                                                                                                                                                                                  |
| <b>Inhalation</b>                          | Remove to fresh air. If breathing is difficult, give oxygen. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                          |
| <b>Most important symptoms and effects</b> | Causes burns by all exposure routes. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation                                                                                                                                                                                   |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                                                                                                                                                   |

## 5. Fire-fighting measures

**Suitable Extinguishing Media** CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

**Unsuitable Extinguishing Media** Water

**Flash Point** No information available  
**Method -** No information available

**Autoignition Temperature** No information available

**Explosion Limits**

**Upper** No data available

**Lower** No data available

**Sensitivity to Mechanical Impact** No information available

**Sensitivity to Static Discharge** No information available

### Specific Hazards Arising from the Chemical

Reacts violently with water.

### Hazardous Combustion Products

Thermal decomposition can lead to release of irritating gases and vapors. Hydrogen chloride gas.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
3

**Flammability**  
0

**Instability**  
1

**Physical hazards**  
W

## 6. Accidental release measures

**Personal Precautions** Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Use personal protective equipment as required.

**Environmental Precautions** See Section 12 for additional Ecological Information. Prevent further leakage or spillage if safe to do so. Prevent product from entering drains.

**Methods for Containment and Clean Up** Provide adequate ventilation. Sweep up and remove immediately. Keep in suitable, closed containers for disposal. Do not expose spill to water. Do not flush to sewer.

## 7. Handling and storage

**Handling** Do not allow contact with water. Do not breathe dust. Do not get in eyes, on skin, or on clothing. Handle under an inert atmosphere. Wear personal protective equipment/face protection.

**Storage.** Keep away from water or moist air. Do not store in metal containers. Keep tightly closed in a dry and cool place. Incompatible Materials. Acids. Strong oxidizing agents. Alcohols.

Amines. .

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component               | ACGIH TLV                                              | OSHA PEL                                                                   | NIOSH                                                                                | Mexico OEL (TWA)                                       |
|-------------------------|--------------------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------|
| Zirconium tetrachloride | TWA: 5 mg/m <sup>3</sup><br>STEL: 10 mg/m <sup>3</sup> | (Vacated) TWA: 5 mg/m <sup>3</sup><br>(Vacated) STEL: 10 mg/m <sup>3</sup> | IDLH: 25 mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup><br>STEL: 10 mg/m <sup>3</sup> | TWA: 5 mg/m <sup>3</sup><br>STEL: 10 mg/m <sup>3</sup> |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

### Personal Protective Equipment

#### Eye/face Protection

Tight sealing safety goggles. Face protection shield.

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure. Impervious clothing. Chemical resistant apron. Boots. Impervious gloves.

#### Respiratory Protection

Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

#### Recommended Filter type:

Particulates filter conforming to EN 143.

#### Hygiene Measures

Keep away from food, drink and animal feeding stuffs. When using do not eat, drink or smoke. Contaminated work clothing should not be allowed out of the workplace. Provide regular cleaning of equipment, work area and clothing. Avoid contact with skin, eyes or clothing. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wear suitable gloves and eye/face protection.

## 9. Physical and chemical properties

|                                        |                          |
|----------------------------------------|--------------------------|
| Physical State                         | Powder                   |
| Appearance                             | White                    |
| Odor                                   | Acidic                   |
| Odor Threshold                         | No information available |
| pH                                     | No information available |
| Melting Point/Range                    | 437 °C / 818.6 °F        |
| Boiling Point/Range                    | No information available |
| Flash Point                            | No information available |
| Evaporation Rate                       | Not applicable           |
| Flammability (solid,gas)               | No information available |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | 1.3 hPa @ 190 °C         |
| Vapor Density                          | Not applicable           |
| Specific Gravity                       | No information available |
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |

|                   |                    |
|-------------------|--------------------|
| Viscosity         | Not applicable     |
| Molecular Formula | Cl <sub>4</sub> Zr |
| Molecular Weight  | 233.03             |

## 10. Stability and reactivity

|                                  |                                                                                                 |
|----------------------------------|-------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | Yes                                                                                             |
| Stability                        | Hygroscopic.                                                                                    |
| Conditions to Avoid              | Exposure to moist air or water.                                                                 |
| Incompatible Materials           | Acids, Strong oxidizing agents, Alcohols, Amines,                                               |
| Hazardous Decomposition Products | Thermal decomposition can lead to release of irritating gases and vapors, Hydrogen chloride gas |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                        |
| Hazardous Reactions              | Reacts violently with water. Corrosive to metals.                                               |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

Oral LD50 Category 4. ATE = 300 - 2000 mg/kg.

#### Component Information

| Component               | LD50 Oral                 | LD50 Dermal | LC50 Inhalation |
|-------------------------|---------------------------|-------------|-----------------|
| Zirconium tetrachloride | LD50 = 1688 mg/kg ( Rat ) | Not listed  | Not listed      |

Toxicologically Synergistic Products No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | Causes burns by all exposure routes                                                      |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component               | CAS No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-------------------------|------------|------------|------------|------------|------------|------------|
| Zirconium tetrachloride | 10026-11-6 | Not listed | Not listed | Not listed | Not listed | Not listed |

Mutagenic Effects No information available

Reproductive Effects No information available.

Developmental Effects No information available.

Teratogenicity No information available.

STOT - single exposure None known

STOT - repeated exposure None known

Aspiration hazard No information available

Symptoms / effects, both acute and delayed Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

Endocrine Disruptor Information No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

**Persistence and Degradability** Decomposes in contact with water. Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** No information available.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No UN2503  
 Proper Shipping Name ZIRCONIUM TETRACHLORIDE  
 Hazard Class 8  
 Packing Group III

### TDG

UN-No UN2503  
 Proper Shipping Name ZIRCONIUM TETRACHLORIDE  
 Hazard Class 8  
 Packing Group III

### IATA

UN-No UN2503  
 Proper Shipping Name ZIRCONIUM TETRACHLORIDE  
 Hazard Class 8  
 Packing Group III

### IMDG/IMO

UN-No UN2503  
 Proper Shipping Name ZIRCONIUM TETRACHLORIDE  
 Hazard Class 8  
 Packing Group III

## 15. Regulatory information

### United States of America Inventory

| Component               | CAS No     | TSCA | TSCA Inventory notification -<br>Active-Inactive | TSCA - EPA Regulatory<br>Flags |
|-------------------------|------------|------|--------------------------------------------------|--------------------------------|
| Zirconium tetrachloride | 10026-11-6 | X    | ACTIVE                                           | -                              |

### **Legend:**

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b)** - Notices of Export

Not applicable

### International Inventories

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component               | CAS No     | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|-------------------------|------------|-----|------|-----------|-------|------|------|------|-------|----------|
| Zirconium tetrachloride | 10026-11-6 | -   | X    | 233-058-2 | X     | X    | X    | X    | X     | KE-35654 |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### U.S. Federal Regulations

#### SARA 313

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372

#### SARA 311/312 Hazard Categories

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

#### CWA (Clean Water Act)

| Component               | CWA - Hazardous Substances | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants |
|-------------------------|----------------------------|-----------------------------|------------------------|---------------------------|
| Zirconium tetrachloride | X                          | 5000 lb                     | -                      | -                         |

Clean Air Act Not applicable

OSHA - Occupational Safety and Health Administration Not applicable

#### CERCLA

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355).

| Component               | Hazardous Substances RQs | CERCLA Extremely Hazardous Substances RQs | SARA Reportable Quantity (RQ) |
|-------------------------|--------------------------|-------------------------------------------|-------------------------------|
| Zirconium tetrachloride | 5000 lb                  | -                                         | 5000 lb<br>2270 kg            |

California Proposition 65 This product does not contain any Proposition 65 chemicals.

### U.S. State Right-to-Know Regulations

| Component               | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|-------------------------|---------------|------------|--------------|----------|--------------|
| Zirconium tetrachloride | X             | X          | X            | -        | X            |

#### U.S. Department of Transportation

Reportable Quantity (RQ): Y  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant N

U.S. Department of Homeland Security This product does not contain any DHS chemicals.

### Other International Regulations

Mexico - Grade No information available

## Authorisation/Restrictions according to EU REACH

| Component               | CAS No     | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------------------|------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Zirconium tetrachloride | 10026-11-6 | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)            | -                                                                                                     |

## REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component               | CAS No     | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-------------------------|------------|----------|------------------------------|---------------------------|--------------------------------------------|
| Zirconium tetrachloride | 10026-11-6 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

## Contains component(s) that meet a 'definition' of per &amp; poly fluoroalkyl substance (PFAS)?

Not applicable

## Other International Regulations

| Component               | CAS No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Zirconium tetrachloride | 10026-11-6 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

## Prepared By

Health, Safety and Environmental Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

## Creation Date

06-Nov-2010

## Revision Date

26-Mar-2024

## Print Date

26-Mar-2024

## Revision Summary

New emergency telephone response service provider.

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**